InsectAI datathon! 2026

AARHUS UNIVERSITY
DEPARTMENT OF ECOSCIENCE

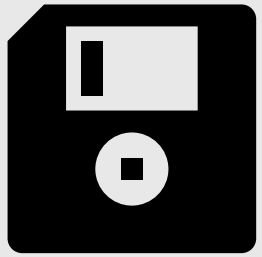
JAMIE ALISON (JALISON@ECOS.AU.DK)
TENURE-TRACK RESEARCHER

GOALS & OUTCOMES

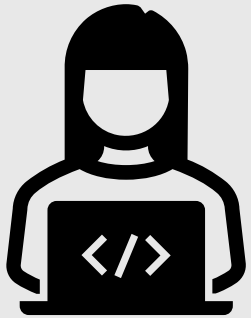
1. Minidatasets to browse on github
2. A refined standard for InsectAI (report 2)
3. Communicating the experience
4. Scripts to process/standardize data

MORE SPECIFICALLY, OUR SCOPE IS...

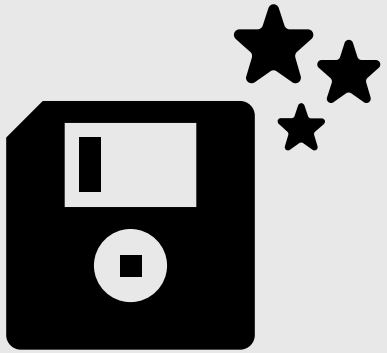
1



RAW DATA

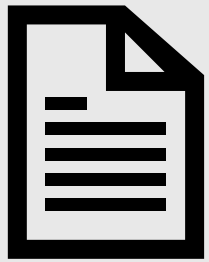


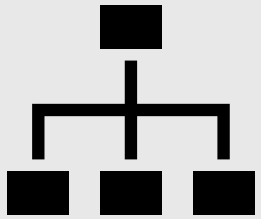
CODE



STANDARDIZED DATA

2



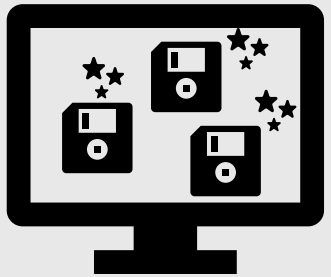


ISSUE LIST FOR THE STANDARD

PROPOSED CHANGES

3

GBIF

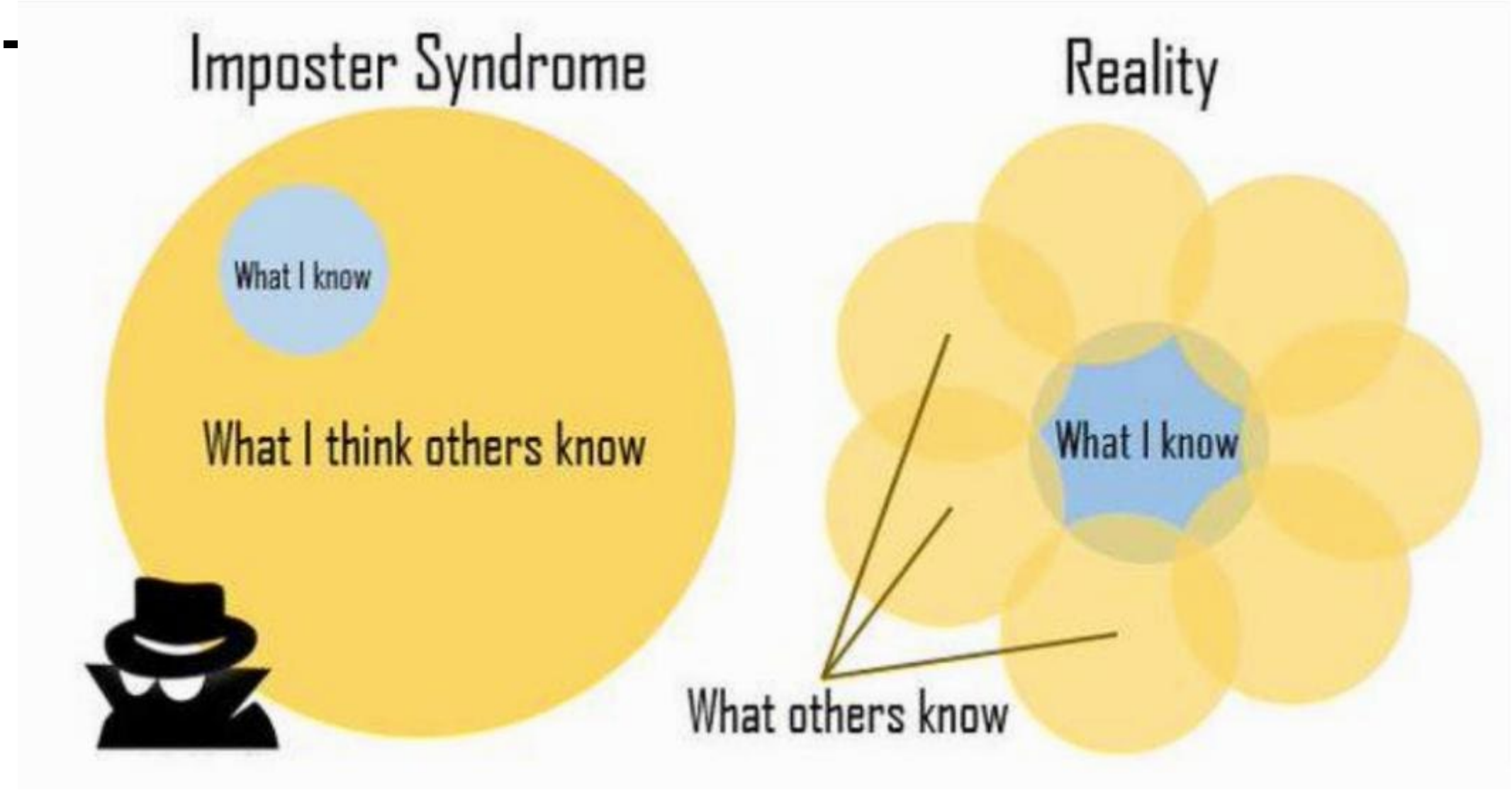


DATA PUBLICATION
COLLATION
BROWSING

TOOLS AND "MEGA-DATASETS"

MOST IMPORTANTLY...

TO THAT END...


Imposter Syndrome
What I know
What I think others know
Reality
What I know
What others know

INTRODUCTIONS/SHOW & TELL

- One minute to find a picture…
- One minute each to “show & tell”
- What do you want to take away from the datathon?

AGENDA - DISCUSSION

0900 - Welcome, purpose & resources - Jamie

0915 - 1 minute madness introductions

0945 - Camtrap DP introduction - Karolina Kuczkowska

1000 - Tools for standards conversion - Sanne Govaert

1015 - Introducing the problems... The datasets and the dreamteams - All

1030 - Coffee break

1100 - datathon! begins.

1230 - Lunch

1330 - Check in & roundtable: Focus on data mapping

1400 - datathon! Continues.

1530 - Coffee break

1630 - Dear data diary: Quickfire feedback from each group (with vectorAI)

0900 - Check in & roundtable: Focus on standard development

0930 - datathon! Continues & some members "adopt a dataset"

1030 - Coffee break

1230 - Lunch

1330 - Awesome InsectAI suggestions for model db – Jamie/Wouter

1400 - Standards dev: Suggestions for Camtrap DP from bioacoustics - Sanne

1415 - Standards dev: Suggestions for the InsectAI standard - Wouter

1530 - Coffee break

1630 - Dear data diary: Quickfire feedback from each

CAMTRAP DP OR INSECTAI EXTENSION?

- Proposal to start with vanilla Camtrap DP
- Bear in mind some possible issues (discussed shortly)
- Revisit Report 2 and collate issues tomorrow

CAMTRAP DP DEVELOPER INSIGHTS...

Karolina Kuczkowska

DEPLOYING AND USING CAMTRAP DP

Sanne Govaert

KEY CHALLENGES IN INSECTAI STANDARDS

- Revisit some previous discussions
- Themes: Observations vs. detections
- Actions: Own standard, change from within, or both?

CROWDSOURCING PROBLEMS

When we have outputs from multiple models and/or annotators

COUNTING PROBLEMS

When we aren’t sure what timescale constitutes an observation

CROPPING PROBLEMS

When we only want to publish subsets/regions of the source imagery

CONFIDENCE PROBLEMS

When we don’t know if we can trust the taxonomic ID, or if it sufficiently fine

COVERAGE PROBLEMS

We looked for some types/sizes of animal, but ignored others

How do we quantify the negatives?
How do we publish them?

FOR ~5 ISSUES, CAN WE FIND OUT

CROWDSOURCING	Is it common?
COUNTING	Can Camtrap DP deal with it
CROPPING	Can we handle it better?
CONFIDENCE	
COVERAGE	... can we make do?

THE GITHUB – STRUCTURE DISCUSSION

- Readme content – should we aim for specific headings?
- A nice table of the datasets, descriptions and example images?

THE GITHUB – STRUCTURE DISCUSSION

📦 Dataset Subfolders

Each individual dataset is located in a folder named `<DATASET_NAME>` with the following internal structure:

Component	Description
`media/`	Folder containing all images, potentially organized into subdirectories.
`raw_labels/`	The original annotations in their source format (JSON, CSV, TXT, etc.).
`code/`	Folder containing the conversion scripts (Jupyter, R, etc.) used to convert the dataset in raw format to the Camtrap DP standard.
`README.md`	A readme file describing the dataset, its source, and any specific details about the conversion process.
`deployments.csv`	**Generated:** Records of camera/sensor deployments.
`media.csv`	**Generated:** Metadata for all media files.
`observations.csv`	**Generated:** Taxonomic or individual observations.
`datapackage.json`	**Generated:** The metadata descriptor for the data package.

DATASET INTRODUCTIONS...

1. Minidatasets: Max 200mb each
2. Must have labels of some kind
3. Capture the challenges of the full data
4. Raw data can take any form

DATASET LUMPING?

- AMI-type
- Flower-type
- Platform-type
- Any more?

TEAM FORMATION...

- Presentation & communication
- Testing & application
- Coding & development
- Any more?

THE ENDLESS TO DO LIST

- Use the github & feed back
- Use code & feed back
- Write/edit readmes & guides
- Write/edit code
- Work on a presentation slide for the dataset. Other PR?
- Read datasets using OS tools
- Explore the “standardization challenges” (the 5 Cs)
- Try to start the process of pushing a dataset to GBIF
- Try to collate minidatasets
- Plan a paper?